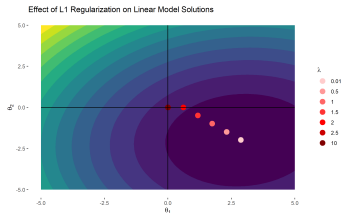
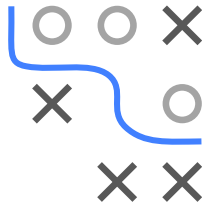


Introduction to Machine Learning

Regularization

Lasso Regression



Learning goals

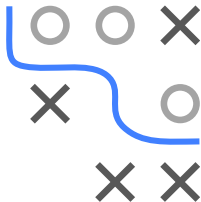
- Lasso regression / L_1 penalty
- Know that lasso selects features
- Support recovery

LASSO REGRESSION I

Another shrinkage method is the so-called **lasso regression** (least absolute shrinkage and selection operator), which uses an L_1 penalty on θ :

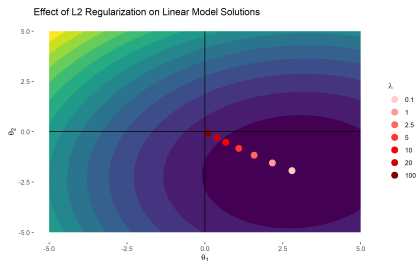
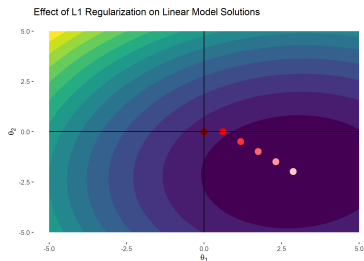
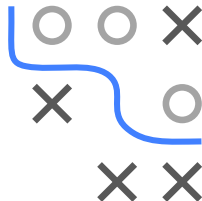
$$\begin{aligned}\hat{\theta}_{\text{lasso}} &= \arg \min_{\theta} \sum_{i=1}^n \left(y^{(i)} - \theta^T \mathbf{x}^{(i)} \right)^2 + \lambda \sum_{j=1}^p |\theta_j| \\ &= \arg \min_{\theta} (\mathbf{y} - \mathbf{X}\theta)^T (\mathbf{y} - \mathbf{X}\theta) + \lambda \|\theta\|_1\end{aligned}$$

Optimization is much harder now. $\mathcal{R}_{\text{reg}}(\theta)$ is still convex, but in general there is no analytical solution and it is non-differentiable.



LASSO REGRESSION II

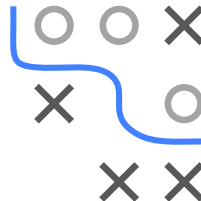
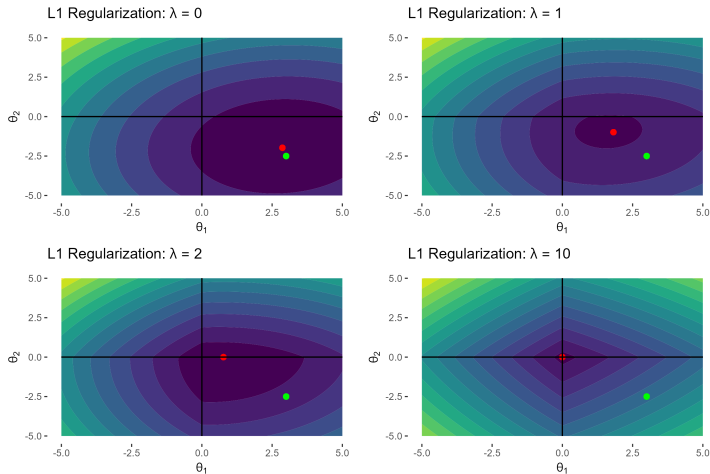
Let $y = 3x_1 - 2x_2 + \epsilon$, $\epsilon \sim N(0, 1)$. The true minimizer is $\theta^* = (3, -2)^T$. LHS = $L1$ regularization; RHS = $L2$



With increasing regularization, $\hat{\theta}_{lasso}$ is pulled back to the origin, but takes a different “route”. θ_2 eventually becomes 0!

LASSO REGRESSION III

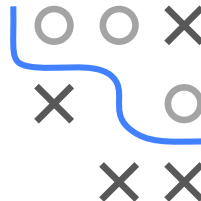
Contours of regularized objective for different λ values.



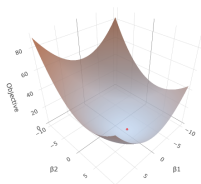
Green = true minimizer of the unreg. objective and red = lasso solution.

LASSO REGRESSION IV

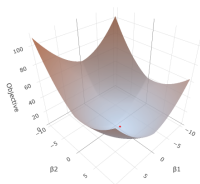
Regularized empirical risk $\mathcal{R}_{\text{reg}}(\theta_1, \theta_2)$ using squared loss for $\lambda \uparrow$. L_1 penalty makes non-smooth kinks at coordinate axes more pronounced, while L_2 penalty warps $\mathcal{R}_{\text{reg}}$ toward a “basin” (elliptic paraboloid).



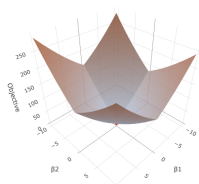
Regularization: L1 λ : 0



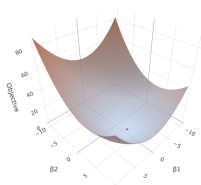
Regularization: L1 λ : 1



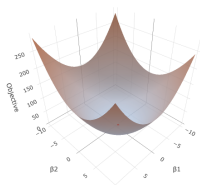
Regularization: L1 λ : 10



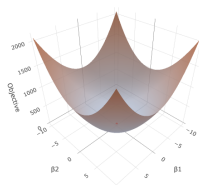
Regularization: L2 λ : 0



Regularization: L2 λ : 1



Regularization: L2 λ : 10

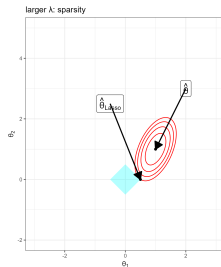
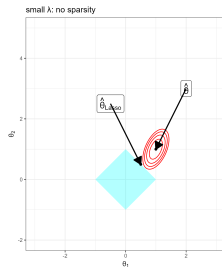
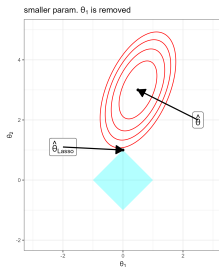
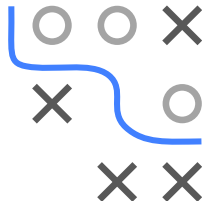


LASSO REGRESSION V

We can also rewrite this as a constrained optimization problem. The penalty results in the constrained region to look like a diamond shape.

$$\min_{\theta} \sum_{i=1}^n \left(y^{(i)} - f(\mathbf{x}^{(i)} | \theta) \right)^2 \text{ subject to: } \|\theta\|_1 \leq t$$

The kinks in $L1$ enforce sparse solutions because “the loss contours first hit the sharp corners of the constraint” at coordinate axes where (some) entries are zero.



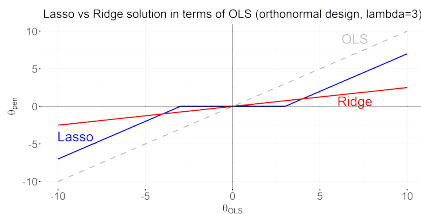
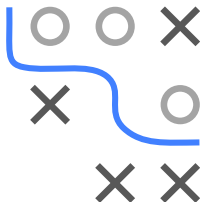
L1 AND L2 REG. WITH ORTHONORMAL DESIGN I

For special case of orthonormal design $\mathbf{X}^T \mathbf{X} = \mathbf{I}$ we can derive a closed-form solution in terms of $\hat{\theta}_{OLS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{X}^T \mathbf{y}$:

$$\hat{\theta}_{\text{lasso}} = \text{sign}(\hat{\theta}_{OLS})(|\hat{\theta}_{OLS}| - \lambda)_+ \quad (\text{sparsity})$$

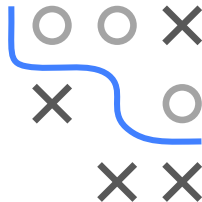
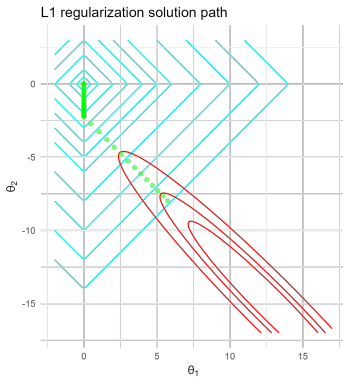
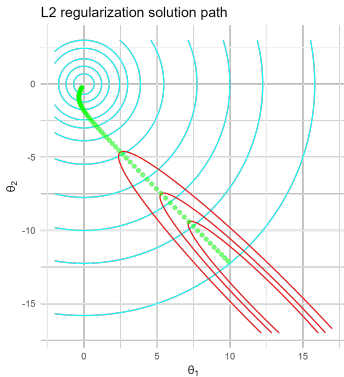
Function $S(\theta, \lambda) := \text{sign}(\theta)(|\theta| - \lambda)_+$ is called **soft thresholding** operator:
For $|\theta| \leq \lambda$ it returns 0, whereas params $|\theta| > \lambda$ are shrunk toward 0 by λ .
Comparing this to $\hat{\theta}_{\text{Ridge}}$ under orthonormal design:

$$\hat{\theta}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} = ((1 + \lambda) \mathbf{I})^{-1} \hat{\theta}_{OLS} = \frac{\hat{\theta}_{OLS}}{1 + \lambda} \quad (\text{no sparsity})$$



COMPARING SOLUTION PATHS FOR $L1/L2$ I

- Ridge results in smooth solution path with non-sparse params
- Lasso induces sparsity, but only for large enough λ



SUPPORT RECOVERY OF LASSO

► Zhao and Yu 2006

When can lasso select true support of θ , i.e., only the non-zero parameters?
Can be formalized as sign-consistency:

$$\mathbb{P}(\text{sign}(\hat{\theta}) = \text{sign}(\theta)) \rightarrow 1 \text{ as } n \rightarrow \infty \quad (\text{where } \text{sign}(0) := 0)$$

Suppose the true DGP given a partition into subvectors $\theta = (\theta_1, \theta_2)$ is

$$Y = X\theta + \varepsilon = X_1\theta_1 + X_2\theta_2 + \varepsilon \text{ with } \varepsilon \sim (0, \sigma^2 I)$$

and only θ_1 is non-zero. Let X_1 denote the $n \times q$ matrix with the relevant features and X_2 the matrix of noise features. It can be shown that $\hat{\theta}_{\text{lasso}}$ is sign consistent under an **irrepresentable condition**:

$$|(\mathbf{X}_2^\top \mathbf{X}_1)(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \text{sign}(\theta_1)| < \mathbf{1} \text{ (element-wise)}$$

In fact, lasso can only be sign-consistent if this condition holds.

Intuitively, the irrelevant variables in X_2 must not be too correlated with (or *representable* by) the informative features

► Meinshausen and Yu 2009

